

Optimization

Course bibliography

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How to use this bibliography

- **Main bibliography** (3 books): the starting point for studying the material.
- **Supplementary bibliography** (7 books): for going deeper into a topic; you are not expected to read them all.
- **Course resources**: the lecturer's theses (the first in Portuguese) and the papers on the GLODS, DMS and MultiGLODS methods.
- **Student projects**: internship reports at Tecmic and IST theses that apply course methods to real problems.
- **Other references**: the original sources of the methods and results cited in the slides.
- Each deck ends with a “References and resources” page with 2–4 references on the topic; in parentheses, in grey, the decks where each reference is used.

- Nocedal, J., Wright, S. J., *Numerical Optimization*, 2nd ed., Springer, 2006. (decks 0, R, 2.1, 2.2, 2.6, 3.1, 3.3, 3.3.1, 3.3.2, 3.3.3, 4.1, 4.2, 4.3.1, 4.3.2)
- Conn, A. R., Scheinberg, K., Vicente, L. N., *Introduction to Derivative-Free Optimization*, SIAM (MPS-SIAM Series on Optimization 8), 2009. (decks 0, 3.2, 3.2.1, 3.2.2, 3.2.3, 3.2.5, 5.1)
- Miettinen, K., *Nonlinear Multiobjective Optimization*, Kluwer Academic Publishers, 1999. (decks 0, 6.1, 6.2, 6.3, 6.4)

- Arora, J. S., *Introduction to Optimum Design*, 3rd ed., Academic Press, 2011. (decks 0, 1.1, 1.2, 3.2.1)
- Deb, K., *Optimization for Engineering Design: Algorithms and Examples*, 2nd ed., PHI Learning, 2012. (decks 0, 2.1, 2.2, 2.3, 2.4, 2.5, 2.6, 3.2.2, 3.2.4, 4.2, 4.3.1, 4.3.2)
- Deb, K., *Multi-Objective Optimization Using Evolutionary Algorithms*, Wiley, 2001. (decks 0, 6.6)
- Dennis, J. E., Jr., Schnabel, R. B., *Numerical Methods for Unconstrained Optimization and Nonlinear Equations*, SIAM (Classics in Applied Mathematics 16), 1996. (decks 0, R, 2.4, 2.5, 3.1, 3.3, 3.3.2)
- Fletcher, R., *Practical Methods of Optimization*, 2nd ed., Wiley, 1987. (decks 0, 2.3, 3.3.1, 4.1, 4.2)
- Horst, R., Tuy, H., *Global Optimization: Deterministic Approaches*, 3rd ed., Springer, 1996. (decks 0, 5.1)
- Jahn, J., *Vector Optimization: Theory, Applications, and Extensions*, 2nd ed., Springer, 2011. (decks 0, 6.2, 6.3)

Course resources: lecturer's theses and papers (1/2)

- Madeira, J. F. A., *Optimização multi-objectivo da topologia de estruturas com algoritmos genéticos*, PhD thesis in Mechanical Engineering, IST, Universidade Técnica de Lisboa, 2004. (decks 0, 5.3, 6.6)
- Madeira, J. F. A., *Multiobjective and global optimization with directional direct search*, PhD thesis in Mathematics, IST, Universidade de Lisboa, 2019. (decks 0, 5.1, 6.5)
- Custódio, A. L., Madeira, J. F. A., *GLODS: Global and Local Optimization using Direct Search*, Journal of Global Optimization 62(1):1–28, 2015. DOI 10.1007/s10898-014-0224-9 (decks 0, 5.1)
- Custódio, A. L., Madeira, J. F. A., *MultiGLODS: global and local multiobjective optimization using direct search*, Journal of Global Optimization 72(2):323–345, 2018. DOI 10.1007/s10898-018-0618-1 (decks 0, 5.1, 6.5)

Course resources: lecturer's theses and papers (2/2)

- Custódio, A. L., Madeira, J. F. A., Vaz, A. I. F., Vicente, L. N., *Direct multisearch for multiobjective optimization*, SIAM Journal on Optimization 21(3):1109–1140, 2011. DOI 10.1137/10079731X (decks 0, 6.5)
- Madeira, J. F. A., *GLODS-SI: scale-invariant global–local direct search for engineering design optimization*, Journal of Computational Design and Engineering 13(6):50–70, 2026. DOI 10.1093/jcde/qwag049 (decks 5.1)
- Madeira, J. F. A., Araújo, A. L., Mota Soares, C. M., Mota Soares, C. A., Ferreira, A. J. M., *Multiobjective design of viscoelastic laminated composite sandwich panels*, Composites Part B: Engineering 77:391–401, 2015. DOI 10.1016/j.compositesb.2015.03.025 (decks 6.1, 6.5)
- Monte, S. M. C., Infante, V., Madeira, J. F. A., Moleiro, F., *Optimization of fibers orientation in a composite specimen*, Mechanics of Advanced Materials and Structures 24(5):410–416, 2017. DOI 10.1080/15376494.2016.1191099 (decks 5.1)

Student projects: internships at Tecmic

Internship reports (co-)supervised by the lecturer, with real operational data (Tecmic, Carris). Reports in Portuguese; titles in the original language.

- Santos, J. R., *Otimização da alocação de serviços de transporte*, internship report, BSc in Applied Mathematics for Technology and Business (LMATE), ISEL, 2022 — simulated annealing. (decks 1.2, 5.2)
- Costa, M. O., *Otimização da alocação de serviços de transporte e logística*, internship report, BSc LMATE, ISEL, 2023 — simulated annealing. (decks 1.2, 5.2)
- Jin, C., *Planeamento de afetação de veículos de transporte público*, internship report, MSc in Applied Mathematics for Industry (MAI), ISEL, 2024 — simulated annealing and NSGA-II. (decks 1.2, 5.2, 6.6)
- Santos, G., *Planeamento de afetação de veículos de transporte público*, internship report, BSc LMATE, ISEL, 2025 — NSGA-II. (decks 1.2, 6.6)
- Macau, R. O., *Alocação dinâmica de serviços*, internship report, BSc LMATE, ISEL, 2026 — greedy, tabu search and simulated annealing. (decks 1.2, 5.2)

IST theses using the course methods (1/3)

Theses (co-)supervised by the lecturer; titles and abstracts on IST's Fénix. **GLODS** and **DMS** in engineering design.

GLODS (5.1)

- Figueiredo, I. M. P., *Previsão do comportamento mecânico e otimização de placas em material compósito híbrido*, MSc Aerospace Eng., IST, 2016 — GLODS, ply orientations (discrete) (deck 5.1)
- Mareco, J. M. C. S., *Otimização de reforços em painéis planos com vigas standard utilizando métodos de otimização discreta para critérios de rigidez*, MSc Aerospace Eng., IST, 2016 — GLODS, beam position and type (discrete) (deck 5.1)
- Vieira, R. M. G. A., *High speed train pantograph models identification*, MSc Mechanical Eng., IST, 2016 — model identification: GLODS and DMS (deck 1.1, 5.1)
- Luís, B. L., *Optimization of passive shunted damping configurations for displacement amplitude reduction of a plate*, MSc Mechanical Eng., IST, 2022 — GLODS and SA on the same problem (deck 5.1, 5.2)
- Magalhães, P. A. P., *Multimodal optimization of piezoelectric patch placement and shunt circuit design for enhanced structural vibration control*, MSc Mechanical Eng., IST, 2025 — GLODS and DMS; full text available from 14/10/2026 (deck 5.1, 6.5)

DMS (6.5)

- Santos, A. M. A., *Dynamic analysis and design of impact attenuator structures for a Formula Student prototype*, MSc Mechanical Eng., IST, 2017 — DMS, mass and maximum acceleration (deck 6.5)
- Castro, J. M., Araújo, A. L., Madeira, J. F. A., *Design and multi-objective optimization of a composite impact attenuator for a Formula Student car*, in *Advances in Engineering Materials, Structures and Systems*, CRC Press, 2019, pp. 498–503 — DMS (from Castro's 2018 thesis) (deck 6.5)
- Caldeira, P. S. A., *Multi-objective topology optimization for bi-dimensional structures*, MSc Aerospace Eng., IST, 2018 — DMS with 900 variables (deck 6.5)
- Ferreira, J. F. M. A., *Structural analysis and optimization of a UAV wing*, MSc Mechanical Eng., IST, 2018 — DMS, mass and deflection (deck 6.5)
- Aguincha, F. M. C., *Otimização multi-objetivo de placas compósitas curvas*, MSc Mechanical Eng., IST, 2019 — DMS, displacement, mass and cost (deck 6.5)
- Dionísio, J. F. M., *Structural optimization of a landing gear*, MSc Aerospace Eng., IST, 2019 — DMS, discrete variables and material (categorical) (deck 6.5)

DMS (6.5), continued

- Fontana, M., *Optimization of composite structural components for energy absorption*, MSc Aerospace Eng., IST, 2019 — DMS, up to 4 objectives (deck 6.5)
- Cotrim, B. R., *Optimal passive shunted damping configurations for noise reduction in sandwich panels*, MSc Mechanical Eng., IST, 2021 — DMS with 7 objectives (deck 6.5)
- Pereira, M. M., *Optimization of wind farm operation towards life-time extension*, MSc Mechanical Eng., IST, 2024 — DMS, power and fatigue (deck 6.5)
- Costa, N. A. R., *Towards next-generation acoustic metamaterials: design and optimisation of sandwich, perforated, and resonant structures*, MSc Aerospace Eng., IST, 2025 — DMS, materials (categorical) (deck 6.5)
- Cunha, M. A. C. A., *Multiobjective optimization of auxetic aerospace cellular structures using a Python implementation of Direct Multisearch*, MSc Aerospace Eng., IST, 2026 — DMS in Python; full text available from 30/04/2027 (deck 6.5)
- Luis, N. F., *Optimal design of piezoelectric sensors and actuators for the attenuation of vibrations in multilaminated structures*, PhD Aerospace Eng., IST, 2016 — DMS with 3 objectives; abstract public (deck 6.5)

Other references (Al-Baali–Chankong)

- Al-Baali, M., *Descent property and global convergence of the Fletcher–Reeves method with inexact line search*, IMA Journal of Numerical Analysis 5(1):121–124, 1985. DOI 10.1093/imanum/5.1.121 (decks 3.3.3)
- Audet, C., Dennis Jr., J. E., *Analysis of generalized pattern searches*, SIAM Journal on Optimization 13(3):889–903, 2003. DOI 10.1137/S1052623400378742 (decks 5.1)
- Audet, C., Dennis Jr., J. E., *Mesh adaptive direct search algorithms for constrained optimization*, SIAM Journal on Optimization 17(1):188–217, 2006. DOI 10.1137/040603371 (decks 5.1)
- Box, G. E. P., *Evolutionary operation: a method for increasing industrial productivity*, Journal of the Royal Statistical Society, Series C (Applied Statistics) 6(2):81–101, 1957. DOI 10.2307/2985505 (decks 3.2.4)
- Box, M. J., *A new method of constrained optimization and a comparison with other methods*, The Computer Journal 8(1):42–52, 1965. DOI 10.1093/comjnl/8.1.42 (decks 3.2.4)
- Chankong, V., Haimes, Y. Y., *Multiobjective Decision Making: Theory and Methodology*, North-Holland, 1983. (decks 6.4)

Other references (Deb–Geman)

- Deb, K., *An efficient constraint handling method for genetic algorithms*, Computer Methods in Applied Mechanics and Engineering 186(2–4):311–338, 2000. DOI 10.1016/S0045-7825(99)00389-8 (decks 4.3.2, 6.6)
- Deb, K., Jain, H., *An evolutionary many-objective optimization algorithm using reference-point-based nondominated sorting approach, Part I: solving problems with box constraints*, IEEE Transactions on Evolutionary Computation 18(4):577–601, 2014. DOI 10.1109/TEVC.2013.2281535 (decks 6.6)
- Deb, K., Pratap, A., Agarwal, S., Meyarivan, T., *A fast and elitist multiobjective genetic algorithm: NSGA-II*, IEEE Transactions on Evolutionary Computation 6(2):182–197, 2002. DOI 10.1109/4235.996017 (decks 6.6)
- Fletcher, R., Reeves, C. M., *Function minimization by conjugate gradients*, The Computer Journal 7(2):149–154, 1964. DOI 10.1093/comjnl/7.2.149 (decks 3.3.3)
- Gass, S., Saaty, T., *The computational algorithm for the parametric objective function*, Naval Research Logistics Quarterly 2(1–2):39–45, 1955. DOI 10.1002/nav.3800020106 (decks 6.3)
- Geman, S., Geman, D., *Stochastic relaxation, Gibbs distributions, and the Bayesian restoration of images*, IEEE Transactions on Pattern Analysis and Machine Intelligence PAMI-6(6):721–741, 1984. DOI 10.1109/TPAMI.1984.4767596 (decks 5.2)

Other references (Geoffrion–Holland)

- Geoffrion, A. M., *Proper efficiency and the theory of vector maximization*, Journal of Mathematical Analysis and Applications 22(3):618–630, 1968. DOI 10.1016/0022-247X(68)90201-1 (decks 6.3)
- Goldberg, D. E., *Genetic Algorithms in Search, Optimization, and Machine Learning*, Addison-Wesley, 1989. (decks 5.3)
- Haimes, Y. Y., Lasdon, L. S., Wismer, D. A., *On a bicriterion formulation of the problems of integrated system identification and system optimization*, IEEE Transactions on Systems, Man, and Cybernetics SMC-1(3):296–297, 1971. DOI 10.1109/TSMC.1971.4308298 (decks 6.4)
- Hajek, B., *Cooling schedules for optimal annealing*, Mathematics of Operations Research 13(2):311–329, 1988. DOI 10.1287/moor.13.2.311 (decks 5.2)
- Hillier, F. S., Lieberman, G. J., *Introduction to Operations Research*, 2024 Release, McGraw-Hill, 2024. (decks 1.1, 5.2)
- Holland, J. H., *Adaptation in Natural and Artificial Systems*, University of Michigan Press, 1975. (decks 5.3)

Other references (Hooke–Metropolis)

- Hooke, R., Jeeves, T. A., “*Direct search*” solution of numerical and statistical problems, *Journal of the ACM* 8(2):212–229, 1961. DOI 10.1145/321062.321069 (decks 3.2.5)
- Kirkpatrick, S., Gelatt Jr., C. D., Vecchi, M. P., *Optimization by simulated annealing*, *Science* 220(4598):671–680, 1983. DOI 10.1126/science.220.4598.671 (decks 5.2)
- Kolda, T. G., Lewis, R. M., Torczon, V., *Optimization by direct search: new perspectives on some classical and modern methods*, *SIAM Review* 45(3):385–482, 2003. DOI 10.1137/S003614450242889 (decks 3.2, 5.1)
- Markowitz, H., *Portfolio selection*, *The Journal of Finance* 7(1):77–91, 1952. DOI 10.1111/j.1540-6261.1952.tb01525.x (decks 1.2)
- McKinnon, K. I. M., *Convergence of the Nelder–Mead simplex method to a nonstationary point*, *SIAM Journal on Optimization* 9(1):148–158, 1998. DOI 10.1137/S1052623496303482 (decks 3.2.3, 3.2.5)
- Metropolis, N., Rosenbluth, A. W., Rosenbluth, M. N., Teller, A. H., Teller, E., *Equation of state calculations by fast computing machines*, *The Journal of Chemical Physics* 21(6):1087–1092, 1953. DOI 10.1063/1.1699114 (decks 5.2)

Other references (Nelder–Zitzler)

- Nelder, J. A., Mead, R., *A simplex method for function minimization*, The Computer Journal 7(4):308–313, 1965. DOI 10.1093/comjnl/7.4.308 (decks 3.2.3)
- Powell, M. J. D., *Restart procedures for the conjugate gradient method*, Mathematical Programming 12(1):241–254, 1977. DOI 10.1007/BF01593790 (decks 3.3.3)
- Rudolph, G., *Convergence analysis of canonical genetic algorithms*, IEEE Transactions on Neural Networks 5(1):96–101, 1994. DOI 10.1109/72.265964 (decks 5.3)
- Torczon, V., *On the convergence of pattern search algorithms*, SIAM Journal on Optimization 7(1):1–25, 1997. DOI 10.1137/S1052623493250780 (decks 3.2.5, 5.1)
- Zhang, Q., Li, H., *MOEA/D: a multiobjective evolutionary algorithm based on decomposition*, IEEE Transactions on Evolutionary Computation 11(6):712–731, 2007. DOI 10.1109/TEVC.2007.892759 (decks 6.6)
- Zitzler, E., Laumanns, M., Thiele, L., *SPEA2: Improving the Strength Pareto Evolutionary Algorithm*, TIK-Report 103, ETH Zürich, 2001. (decks 6.6)

- **Course code** (MATLAB and Python): functions and examples that reproduce the numbers in the slides, in the course repository.
- **GLDS** (Custódio and Madeira, 2015): version 0.2, distributed in class; official page `docentes.fct.unl.pt/algb/pages/glods` (latest version: 0.3).
- **DMS** (Custódio, Madeira, Vaz and Vicente, 2011): `www.mat.uc.pt/dms`. In MATLAB, `paretosearch` (since R2018b) is a multiobjective pattern search whose documentation refers to this paper for the convergence analysis.
- **MATLAB**: Optimization Toolbox (`fminbnd`, `fminsearch`, `fminunc`, `fmincon`, `linprog`) and Global Optimization Toolbox (`patternsearch`, `simulannealbnd`, `ga`, `gamultiobj`, `paretosearch`).
- **Python**: SciPy (`scipy.optimize`) and pymoo (NSGA-II).